

Trace Element Geochemistry: Partition Coefficients, Incompatibility & Igneous Differentiation

Comprehensive Geochemistry & Petrology Guide · Authoritative Reference: John D. Winter, Principles of Igneous and Metamorphic Petrology (2nd Ed., Ch. 9)

THEORETICAL FOUNDATION: TRACE ELEMENT FRACTIONATION & PARTITIONING

Distribution Coefficients (D), Batch Melting, Rayleigh Fractional Crystallization & Geochemical Reservoirs

1. The Physics and Chemistry of Trace Element Partitioning

Trace elements are defined as elements present in rocks at concentrations less than $0.1 \text{ wt}\%$ ($< 1000 \text{ ppm}$). Because their abundances are so low, trace elements do not form distinct mineral phases of their own (such as quartz or feldspar); instead, they substitute into the crystal lattices of major rock-forming minerals governed by **Goldschmidt's Rules of Substitution** (matching ionic radius and charge):

THE DISTRIBUTION COEFFICIENT (K_D OR D_i)

$$D_i = \frac{C_{\text{solid}}}{C_{\text{liquid}}}$$

Where:

- C_{solid} = Weight concentration of element i in the crystalline mineral phase.
- C_{liquid} = Weight concentration of element i in the coexisting silicate melt.

For a multi-mineralic rock, the **Bulk Distribution Coefficient (D)** is the weighted average:

$$D = \sum_j W_j \cdot D_j = W_{\text{ol}} D_{\text{ol}} + W_{\text{cpx}} D_{\text{cpx}} + W_{\text{plag}} D_{\text{plag}} + \dots$$

where W_j is the weight fraction of mineral j in the solid residue or cumulate.

Compatible vs. Incompatible Elements:

- Compatible Element ($D > 1$):** Fits comfortably into the mineral lattice sites of common early-crystallizing minerals (e.g., Ni^{2+} into olivine; Cr^{3+} into chromite and clinopyroxene). The element preferentially remains in or enters the solid phase, becoming rapidly depleted in evolving liquids.

• **Incompatible Element ($D < 1$, often $D \ll 1$):** Possesses an ionic radius or valency that is poorly suited for major rock-forming mineral lattices. The element is strongly rejected by solid crystals, partitioning preferentially into the silicate melt ($C_{\text{liquid}} \gg C_{\text{solid}}$).

"Practically all elements distribute themselves unevenly between any two phases at equilibrium. This effect is known as chemical fractionation. An incompatible element is one that has a bulk distribution coefficient D significantly less than 1. Such elements are rejected by the solid and are concentrated into the liquid phase during both partial melting and fractional crystallization."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., 2010), Chapter 9: Chemical Petrology II: Trace Elements and Isotopes, p. 167.

2. Textbook Evidence: Mineral/Melt Partition Coefficients (D -Values)

Table 1 from Winter provides authentic measured partition coefficients for key trace elements across the common rock-forming minerals in basaltic to andesitic magmatic systems:

TABLE 1 Partition Coefficients (C_S/C_L) for Some Commonly Used Trace Elements in Basaltic and Andesitic Rocks							
	Olivine	Opx	Cpx	Garnet	Plag	Amph	Magnetite
Rb	0.010	0.022	0.031	0.042	0.071	0.29	
Sr	0.014	0.040	0.060	0.012	1.830	0.46	
Ba	0.010	0.013	0.026	0.023	0.23	0.42	
Ni	14	5	7	0.955	0.01	6.8	29
Cr	0.70	10	34	1.345	0.01	2.00	7.4
La	0.007	0.03	0.056	0.001	0.148	0.544	2
Ce	0.006	0.02	0.092	0.007	0.082	0.843	2
Nd	0.006	0.03	0.230	0.026	0.055	1.340	2
Sm	0.007	0.05	0.445	0.102	0.039	1.804	1
Eu	0.007	0.05	0.474	0.243	0.1/1.5*	1.557	1
Dy	0.013	0.15	0.582	3.17	0.023	2.024	1
Er	0.026	0.23	0.583	6.56	0.020	1.740	1.5
Yb	0.049	0.34	0.542	11.5	0.023	1.642	1.4
Lu	0.045	0.42	0.506	11.9	0.019	1.563	

Data from Rollinson (1993) and <http://earthref.org>

Figure 1 (Winter Table 1, p. 168): Partition coefficients ($C_{\text{solid}} / C_{\text{liquid}}$) for trace elements in olivine, orthopyroxene, clinopyroxene, garnet, plagioclase, amphibole, and magnetite. Notice that for olivine, D -values for incompatible elements are microscopic: $D_{\text{Rb}} = 0.010$, $D_{\text{Sr}} = 0.014$, $D_{\text{Ba}} = 0.010$, $D_{\text{La}} = 0.007$, $D_{\text{Zr}} = 0.012$. In stark contrast, compatible elements like D_{Ni} in olivine reach $D = 14.0$, and D_{Cr} in clinopyroxene reaches $D = 8.4$.

3. Mathematical Behavior During Melting and Crystallization

Trace elements obey rigorous thermodynamic conservation-of-mass equations during magmatic evolution:

BATCH MELTING (EQUILIBRIUM PARTIAL MELTING)

$$\frac{C_L}{C_0} = \frac{1}{D(1 - F) + F}$$

Where C_0 is source rock concentration, C_L is melt concentration, and F is melt fraction ($0 \leq F \leq 1$).

- For a highly incompatible element ($D \rightarrow 0$):

$$\frac{C_L}{C_0} \approx \frac{1}{F}$$

Physical significance: At small degrees of partial melting (e.g., $F = 0.01 = 1\%$), incompatible elements are concentrated in the first melt by a factor of $1/0.01 = \mathbf{100\times}$ relative to the mantle source!

RAYLEIGH FRACTIONAL CRYSTALLIZATION

$$\frac{C_L}{C_0} = F^{(D - 1)}$$

Where F is the fraction of liquid remaining ($1 \rightarrow 0$).

- For an incompatible element where $D \ll 1$ (say $D \approx 0$):

$$\frac{C_L}{C_0} \approx F^{-1} = \frac{1}{F}$$

Physical significance: As crystallization proceeds from basalt $\rightarrow$ andesite $\rightarrow$ dacite $\rightarrow$ rhyolite/granite, the fraction of liquid remaining shrinks ($F \rightarrow 0.05$), concentrating incompatible trace elements exponentially in the residual granitic melt!

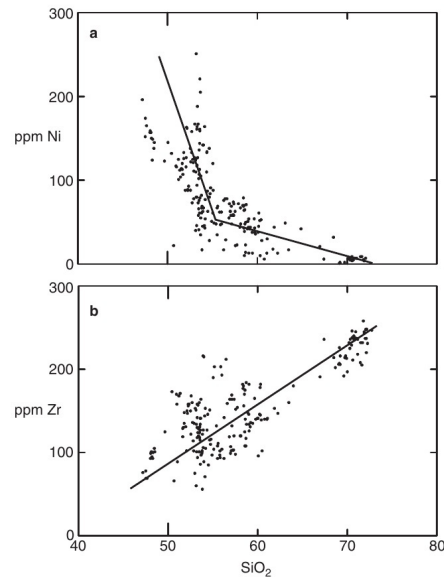


FIGURE 1 Harker diagrams of (a) Ni versus SiO_2 and (b) Zr versus SiO_2 for the Crater Lake suite of volcanic rocks.

Conversely, a trace element with a very small $\overline{D}_i$ is preferentially concentrated in the liquid and will reflect the proportion of liquid at a given state of crystallization or melting because its concentration varies inversely with the amount of liquid diluting it. The plot of Zr versus SiO_2 in Figure 1b is an illustration. Some relatively simple but useful quantitative models have been developed that can treat multiple fractionating phases and various crystallization or melting schemes using a variety of trace elements. We shall discuss a few of these next.

2 MODELS FOR SOLID–MELT PROCESSES

2.1 Batch Melting

Figure 2 (Winter Fig. 1, p. 169): Harker variation diagrams for compatible Ni and incompatible Zr versus SiO_2 for the Crater Lake volcanic suite. Compatible Ni collapses abruptly from 150 ppm to $<10 \text{ ppm}$ during early olivine fractionation. In sharp contrast, incompatible Zr continuously climbs as SiO_2 increases, reaching maximum concentrations in high-silica rhyolites and granites.

4. The Petrogenetic Hierarchy: Dunite $\rightarrow$ Peridotite $\rightarrow$ Basalt $\rightarrow$ Granite

To definitively evaluate Question 18, we trace the abundance of incompatible trace elements across the four candidate rock types:

Rock Type	Dominant Mineralogy	Petrogenetic Origin	Incompatible Trace Element Inventory
Dunite (Ultramafic Cumulate / Residue)	$>90\%$ Olivine ($\sim$ Chromite)	Refractory melt extraction residue or early ultramafic crystal cumulate.	Lowest in Nature: Olivine rejects virtually all incompatible elements ($D_{\text{ol}} < 0.01$). $\text{Rb, Ba, Th, U, Zr, REE}$ are nearly undetectable (< 0.01 to 0.1 ppm).
Peridotite (Mantle Source)	Olivine + Orthopyroxene + Clinopyroxene + Garnet/Spinel	Pristine upper mantle (e.g., Iherzolite) or depleted mantle (harzburgite).	Depleted Baseline: Represents the primitive or depleted mantle reservoir. Serves as the source rock prior to melt extraction (C_0). Incompatible levels are very low (0.1 to 1 Primitive Mantle).
Basalt (Mafic Volcanic)	Calcic Plagioclase + Clinopyroxene ($\sim$ Olivine)	Moderate partial melt of mantle peridotite ($F \approx 10\% - 25\%$).	Moderate Enrichment: Melt captures incompatible elements extracted from the mantle ($C_L \approx 4$ to 10 mantle source), but remains relatively primitive compared to felsic rocks.
Granite ★ Correct Answer	Quartz + K-Feldspar + Sodic Plagioclase + Biotite	Terminal product of extensive fractional crystallization ($F < 10\%$) or low- F crustal partial melting.	Highest Abundance: Extremely enriched in all Large-Ion Lithophile Elements (LILE: Rb, Cs, Ba, K) and High Field Strength Elements (HFSE: $\text{Zr, Hf, Th, U, Nb, Ta}$), reaching 100 to 1000 mantle levels.

"The process of fractional crystallization tends to remove only a few compatible elements from the liquid in significant quantities, and a large number of incompatible, LIL, and non-lithophile elements become concentrated in the latest liquid fraction... Highly evolved granites and pegmatites represent the extreme manifestation of this enrichment."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology*, Chapter 11, p. 219.

5. Classification of Incompatible Trace Elements

Geochemists subdivide incompatible trace elements into two major groups based on the ratio of ionic charge (z) to ionic radius (r), known as **ionic potential (z/r)**:

- **Large-Ion Lithophile (LIL) Elements ($z/r < 2.0$):** Low charge ($+1$ or $+2$) and large ionic radius (> 1.0 Å). Includes Cs^+ , Rb^+ , K^+ , Ba^{2+} , Sr^{2+} , Eu^{2+} . Highly mobile in aqueous fluids; concentrated heavily in continental crust and granites.

- **High Field Strength (HFS) Elements ($Z/r > 2.0$):** High formal charge (+3, +4, +5) and small/intermediate radius. Includes Zr^{4+} , Hf^{4+} , Nb^{5+} , Ta^{5+} , Th^{4+} , U^{4+} , Ti^{4+} , Y^{3+} . Chemically immobile in hydrous fluids; concentrated by magmatic differentiation into felsic magmas and accessory minerals (zircon, monazite, allanite).
- **Rare Earth Elements (REE):** Lanthanides (La^{3+} to Lu^{3+}), where Light REE (LREE, large radii) are significantly more incompatible than Heavy REE (HREE, smaller radii).



EXAM QUESTION & RIGOROUS OPTION-BY-OPTION DISSECTION

DGMS 2019 Examination · Official Question Statement, Verified Answer & Detailed Distracter Analysis

DGMS 2019 · QUESTION 18

Which one of the following rocks is normally expected to have the highest abundance of incompatible trace elements?

- (a) Basalt
- (b) Dunite
- (c) Peridotite
- (d) Granite

Correct Option: (d) Granite



Verified via John D. Winter, Principles of Igneous & Metamorphic Petrology (Ch. 9, pp. 167–172; Ch. 11, p. 219)

INCORRECT Option (a): Basalt

Detailed Refutation: Basalt is a mafic volcanic rock generated by moderate degrees of partial melting ($F \approx 10\%–25\%$) of mantle peridotite. While basalt is substantially enriched in incompatible elements compared to its peridotite mantle source, it is far less differentiated than granite. During subsequent fractional crystallization of basaltic magma, the liquid undergoes an additional $80\%–90\%$ crystal removal, concentrating incompatible elements several orders of magnitude higher into terminal granitic melts.

INCORRECT Option (b): Dunite

Detailed Refutation: Dunite is an ultramafic rock composed of $>90\%$ olivine. Because the olivine crystal lattice strongly excludes incompatible trace elements ($D_{\text{ol}} \approx 0.001–0.01$ for $\text{Rb, Ba, Th, U, REE}$), dunites (whether formed as mantle melting residues or early ultramafic cumulates) have the **lowest** incompatible trace element concentrations of any terrestrial rocks.

INCORRECT Option (c): Peridotite

Detailed Refutation: Peridotite is the primary ultramafic rock of the Earth's upper mantle. It represents the un-melted starting source (primitive mantle) or the depleted residue after basalt extraction (harzburgite). By geochemical definition, the mantle source has baseline trace element concentrations prior to the massive enrichment that occurs during partial melting and prolonged differentiation.

CORRECT ANSWER Option (d): Granite

Detailed Validation: Granite is a highly evolved, felsic, silica-rich continental plutonic rock. It represents either the extreme residual melt produced after prolonged fractional crystallization of more mafic parent magmas ($F \ll 0.1$) or the product of low-degree crustal anatexis. Because incompatible trace elements ($D \ll 1$) are continuously rejected by early-crystallizing minerals (olivine, pyroxene, calcic plagioclase, spinels), they accumulate relentlessly in the residual liquid, reaching their absolute maximum natural abundances in granites, pegmatites, and associated felsic differentiates.



SYNTHESIS & HIGH-YIELD PRACTICE QUESTIONS

Curated Problems Covering Trace Element Modeling, Rayleigh Fractionation, LILE/HFSE & Spidergrams

High-Yield Exam Practice Questions

Practice Q1 (Numerical / Batch Melting): Enrichment Calculation

A perfectly incompatible element ($D = 0$) is present in a mantle peridotite source at a concentration of $C_0 = 1.5 \text{ ppm}$. If the mantle undergoes $F = 3\%$ (0.03) batch partial melting, what is the concentration of this element in the resulting primary basaltic melt?

Solution & Rationale:

Using the batch melting equation: $C_L / C_0 = \frac{1}{D(1-F) + F}$.

Since $D = 0$, $C_L / C_0 = \frac{1}{F} = \frac{1}{0.03} = 33.33$.

$C_L = 1.5 \text{ ppm} \times 33.33 = 50.0 \text{ ppm}$.

The melt is enriched by more than 33-fold relative to the solid mantle source.

Practice Q2 (Rayleigh Fractionation): Compatible Element Depletion

Nickel is strongly compatible in olivine ($D_{\text{Ni}} = 10$). If a basaltic magma undergoes fractional crystallization until only $F = 50\%$ (0.5) of the liquid remains, what fraction of the original nickel remains in the melt?

Solution & Rationale:

Rayleigh equation: $C_L / C_0 = F^{D-1}$.

Here $D - 1 = 10 - 1 = 9$.

$C_L / C_0 = (0.5)^9 = \frac{1}{512} \approx 0.00195 = 0.195\%$.

Over 99.8% of the original nickel is locked into the early-crystallized olivine cumulate, explaining why evolved rocks (andesites, granites) are virtually devoid of nickel.

Practice Q3 (Element Taxonomy): Distinguishing LILE from HFSE

Which of the following trace elements is classified as a High Field Strength Element (HFSE) rather than a Large-Ion Lithophile Element (LILE): (a) Rb , (b) Ba , (c) Zr , (d) Sr ?

Answer: (c) Zr (Zirconium). Zr^{4+} has a high valence charge (+4) and relatively small ionic radius ($0.72 \text{ \AA}$), yielding a high ionic potential ($Z/r = 5.56$). In contrast, Rb^+ , Ba^{2+} , and Sr^{2+} are all low-valence, large-radius LIL elements.

Practice Q4 (Petrogenesis): Incompatible Behavior in Granitic Pegmatites

Why do giant rare-metal pegmatites (containing spodumene, columbite-tantalite, beryl, and uraninite) form almost exclusively as peripheral haloes around granitic plutons?

Answer: Incompatible elements with extreme ionic charges or radii ($\text{Li, Be, B, Nb, Ta, U, Th, REE}$) cannot fit into quartz or feldspar lattices during the crystallization of granite. As the granite body cools, these elements are squeezed into the final, water-saturated residual volatile melt fraction ($F < 0.01$), which injects into surrounding country rocks to crystallize pegmatites.

Practice Q5 (Bulk Partition Coefficient): Mineral Weight Influence

Explain why a rock containing just 3% garnet can behave as a compatible reservoir for Heavy Rare Earth Elements (HREE) like Lutetium (Lu), even when all other minerals in the rock have $D < 1$.

Answer: The partition coefficient for Lu in garnet is exceptionally high ($D_{\text{garnet}}^{\text{Lu}} \approx 12.0$). In the bulk equation $D = \frac{W_{\text{gt}} D_{\text{gt}}}{\sum W_j D_j}$, the garnet term alone contributes $0.03 \times 12.0 = 0.36$. When combined with pyroxenes, the bulk D easily exceeds 1.0, retaining HREE in the garnet-bearing mantle residue.

Practice Q6 (Spidergrams): Primitive Mantle Normalization

Why are trace element concentrations in igneous rocks normalized to Chondrite or Primitive Mantle in "spidergrams" (incompatible element multi-element diagrams)?

Answer: To eliminate the natural "sawtooth" abundance variation caused by the Oddo-Harkins effect (elements with even atomic numbers are cosmochemically more abundant than elements with odd atomic numbers). Normalization produces smooth patterns where slopes and anomalies directly reflect petrogenetic processes.

Practice Q7 (Subduction Signatures): LILE vs. HFSE Decoupling

Why are subduction zone magmas (island arc basalts and calc-alkaline andesites) characteristically enriched in LILE (Rb, Ba, Sr) but depleted in HFSE (Nb, Ta, Ti)?

Answer: LILE are fluid-mobile and are readily transported by aqueous dehydration fluids from the subducting slab into the mantle wedge. HFSE are highly insoluble in aqueous fluids and are retained in the slab within residual rutile (TiO_2), producing the classic negative Nb-Ta and Ti anomalies on arc spidergrams.

Practice Q8 (Mantle Residues): Fertile vs. Depleted Mantle

Between a fertile lherzolite and a depleted harzburgite, which mantle rock has higher incompatible trace element abundances, and why?

Answer: Fertile Iherzolite. Harzburgite has already experienced one or more episodes of basaltic melt extraction, which stripped away its clinopyroxene and incompatible trace elements into the extracted magma, leaving a refractory, depleted residue of olivine + orthopyroxene.

Summary Comparison of Rock Suites and Trace Element Inventories

Rock Type	SiO_2 Content	Compatible Elements (Ni , Cr , Co)	Incompatible Elements (Rb , Ba , Zr , Th , REE)	Degree of Differentiation
Dunite	$\sim 40\text{ wt}\%$	Extreme ($>2000\text{ ppm Ni}$)	Negligible / Depleted ($<0.1\text{ ppm}$)	Refractory residue / Early cumulate
Peridotite	$40\text{--}45\text{ wt}\%$	Very High ($>1500\text{ ppm Ni}$)	Very Low (Mantle source baseline)	Primitive mantle reservoir
Basalt	$45\text{--}52\text{ wt}\%$	Moderate ($50\text{--}250\text{ ppm Ni}$)	Moderate ($10\text{--}50\times$ mantle)	Primary mantle melt
Granite	$70\text{--}77\text{ wt}\%$	Very Low ($<5\text{ ppm Ni}$)	Highest in Nature ($100\text{--}1000\times$ mantle)	Terminal differentiated felsic liquid